Highlights

A Kirkpatrick-based Framework for Integrating Safety and Sustainability in Aviation Maintenance: A Case Study on Waste Reduction and Resource Efficiency

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- Kirkpatrick model applied to aviation MRO over a 12-month case study.
- Training failures decreased 28%; regulatory non-conformances dropped 35%.
- Payback of under twelve months in estimated cost-of-quality savings.
- Framework maps Kirkpatrick levels to AS9100, SMS pillars and ESG indicators.
- Policy guidance proposed for ICAO, FAA, EASA and ANAC oversight regimes.

A Kirkpatrick-based Framework for Integrating Safety and Sustainability in Aviation Maintenance: A Case Study on Waste Reduction and Resource Efficiency

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ABSTRACT

Within high-risk industries such as aviation maintenance, safety and sustainability are intrinsically linked: maintenance errors not only drive accidents but also generate substantial resource waste through rework, scrap and the loss of human capital. This case study operationalises the Kirkpatrick four-level model (Reaction, Learning, Behaviour, Results) over a twelve-month intervention at a multinational aircraft-engine MRO facility and quantifies both safety and sustainability outcomes. Mixed-methods data collection combined post-training questionnaires, on-the-job observations, quality records, warranty and concession costs, customer satisfaction surveys and turnaround time metrics. After implementation, the training-failure rate was 28% lower and the regulatory non-conformance rate 35% lower than baseline (95% CIs in Section 4), with estimated annual quality-cost savings in the range of USD 0.75–1.5 M (base case USD 1.19 M) and a payback period under twelve months across pessimistic-to-optimistic scenarios. The single-site case-study design supports association rather than causal attribution; the results are interpreted accordingly. The intervention also prevented the material and energy waste associated with corrective actions. Customer satisfaction surveys with four major airline customers reported improvements across all five evaluated categories, with mean ratings consistently in the upper range of the five-point scale during the intervention year. We provide a replicable framework for MRO managers and regulators, mapping each Kirkpatrick level onto specific AS9100 clauses, SMS pillars and sustainability indicators. The study supports policy recommendations for integrating training-effectiveness metrics into SMS requirements mandated by ICAO Annex 19, demonstrating that systematic training evaluation can transform technical instruction from a compliance obligation into a strategic lever for cleaner production in resource-intensive aviation operations.

1. Introduction

The global maintenance, repair and overhaul (MRO) market is valued at over USD 90 billion annually and remains a critical determinant of airline operating costs, fleet availability and on-time performance (Zhang, Wang and Chen, 2025; Boyd and Stolzer, 2015). The sector operates under one of the most stringent safety regimes across all engineering domains, where maintenance-attributable errors continue to account for between 12% and 18% of commercial aviation accidents (Boyd and Stolzer, 2015; Federal Aviation Administration, 2024) despite decades of regulatory and technological progress. The economic and environmental consequences of these errors extend well beyond the immediate corrective action: a single warranty claim arising from a defective overhaul has been reported to exceed USD 189,500 in direct material and labour costs alone (Pereira, 2017), and the indirect cost — expressed as material scrap, energy consumed in rework, displaced operational capacity, missed slot-allocation opportunities at congested hubs and reputational damage — has been estimated at three to seven times the direct figure (Zhang et al., 2025). For airlines, every maintenance disruption translates into unit cost pressure and competitive position erosion; for the MRO operator, each event compounds into quality-cost dilution and the slow attrition of preferred-supplier status.

Safety and sustainability in MRO are therefore intrinsically linked. Maintenance errors do not only generate accident risk; they also generate waste. Every component scrapped because of a procedural deviation, every engine returned for rework, every additional litre of jet fuel consumed by a delayed turnaround and every kilowatt-hour of energy invested in corrective actions correspond to a sustainability loss that is rarely reported in isolation but is systematically captured by quality-cost accounting (Chen, Wang, Liu and Zhou, 2023; Yang and Park, 2024). This connection has become particularly relevant under the rising pressure of stakeholder demand for environmental, social and governance (ESG) disclosures from airlines and their supply chain (Sousa, Lima and Costa, 2025).

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Within this context, the technical training of maintenance personnel emerges as a high-leverage intervention. Training is mandated by both the civil aviation regulator (FAA, EASA, ANAC and other authorities under ICAO Annex 19; Federal Aviation Administration, 2024; European Union Aviation Safety Agency, 2025) and the industry quality standard for aerospace, AS9100 (Oschman, 2017). Yet, in many organisations training remains evaluated only at the level of attendance, written tests or post-class satisfaction surveys — a posture that limits training to a compliance obligation and ignores its strategic potential to reduce waste and reinforce safety culture (O'Connor, Flin and Sennett, 2024; Nakamura, Sato and Tanaka, 2024).

The Kirkpatrick four-level model (Kirkpatrick and Kirkpatrick, 2010; Patel, Kumar and Singh, 2023; Fernández-Delgado and Cernadas, 2024) provides a well-established framework to evaluate training effectiveness across Reaction, Learning, Behaviour and Results dimensions. Although broadly cited, its operationalisation in regulated and high-reliability environments such as aviation MRO remains underexplored. Recent reviews suggest that fewer than 20% of published applications go beyond Level 2 (Learning) in any sustained way, and applications that explicitly connect Level 4 (Results) with sustainability or cleaner-production outcomes are virtually absent (O'Reilly, Murphy and Doyle, 2024; Peterson, 2025).

1.1. Research questions

To address this gap, this study investigates the following research questions (RQs):

RQ1. How can the Kirkpatrick four-level model be operationalised within the training process of an AS9100-certified aircraft-engine MRO facility so that it evaluates not only safety and regulatory compliance but also resource efficiency and waste reduction?

RQ2. What measurable changes in safety performance (training failures, regulatory non-conformances) and sustainability indicators (quality costs, customer satisfaction, turnaround time) result from such an implementation over a twelve-month horizon?

RQ3. Which managerial and policy interventions can be derived from the case so that the framework becomes replicable across MRO organisations and useful as guidance for civil aviation authorities and industry bodies?

1.2. Contribution and structure

To the best of the authors' knowledge, the contribution of this paper is threefold: *(a)* it jointly maps Kirkpatrick's four levels to specific AS9100 clauses, SMS pillars and ESG indicators in an aerospace maintenance setting, *and (b)* it reports twelve-month longitudinal operational outcomes, including a monetised cost-of-quality estimate, from a single AS9100-certified engine MRO facility. While prior work has applied Kirkpatrick to aviation training in isolation and has discussed sustainability in MRO separately, the joint mapping with longitudinal monetised outcomes in this regulated setting is, to our knowledge, not yet documented in the peer-reviewed literature. The third contribution is normative: managerial implications and policy recommendations directly addressed to MRO managers, airline quality departments and aviation regulators are derived from the case. The conceptual mapping is summarised in Fig. 1.

The remainder of the paper is organised as follows. Section 2 reviews the relevant literature on Kirkpatrick, aviation maintenance training and the safety-sustainability nexus. Section 3 describes the case-study design and data collection. Section 4 presents the results, including the regulatory mapping, the redesigned training process and the operational outcomes. Section 5 discusses the findings against the literature. Section 6 provides managerial implications, including an economic analysis. Section 7 formulates policy recommendations. Section 8 concludes, and Section 8.4 synthesises limitations and future research.

2. Literature Review

This section examines four bodies of literature that converge in the present study: (i) the Kirkpatrick model and its modern critiques; (ii) aviation maintenance training under regulatory pressure; (iii) human factors and socio-technical perspectives on maintenance error; and (iv) technological and economic enablers for next-generation evaluation. Section 2.6 synthesises the gap.

2.1. Kirkpatrick's model: theoretical evolution and modern critiques

Originally introduced in 1959 and consolidated in subsequent revisions, the four-level model of training evaluation (Kirkpatrick and Kirkpatrick, 2010) remains the most cited framework worldwide for assessing learning interventions. The levels are widely understood as *Reaction* (participant satisfaction), *Learning* (knowledge or skill acquisition), *Behaviour* (transfer of training to the workplace) and *Results* (organisational outcomes). The simplicity of the model explains its longevity, but it has also drawn substantial criticism for treating the four levels as a linear causal chain (Patel et al., 2023; Fernández-Delgado and Cernadas, 2024).

Contemporary work has refined the model along three axes. First, attribution methods such as SHAP-based causal analysis and quasi-experimental design have been proposed to isolate the unique contribution of training when many other variables change simultaneously (Fernández-Delgado and Cernadas, 2024). Second, predictive extensions have been developed to

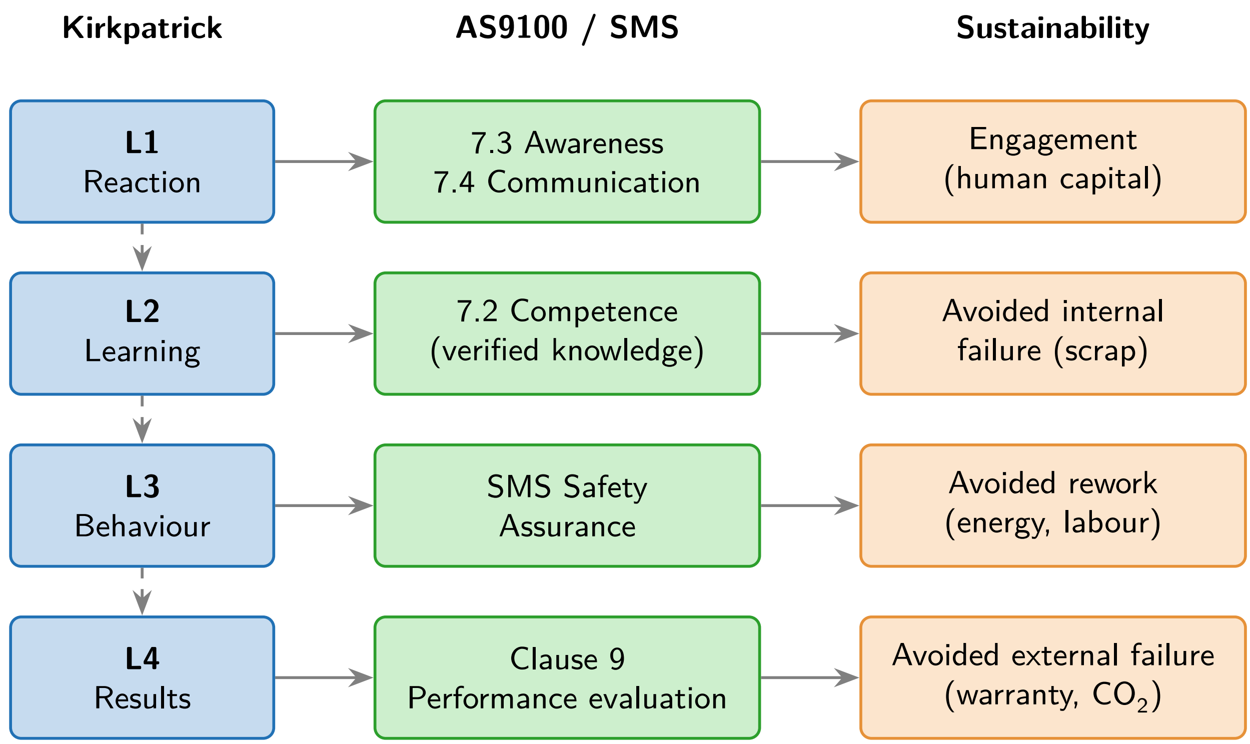

Figure 1: Conceptual model linking Kirkpatrick's four levels to AS9100 clauses, SMS pillars and sustainability outcomes. Vertical dashed arrows represent the progression of the Kirkpatrick model; horizontal arrows represent the mapping proposed in this study.

forecast Level 3 and Level 4 outcomes from Level 1 and Level 2 signals, enabling more proactive management (Peterson, 2025). Third, integrative variants have linked Kirkpatrick to safety culture, human-capital metrics and sustainability indicators (Sousa et al., 2025; Yang and Park, 2024). Despite these advances, empirical applications in regulated heavy-asset sectors remain limited.

2.2. Aviation maintenance training: regulatory imperatives and empirical shortfalls

Aviation MRO is regulated by a multi-level architecture that originates in the ICAO Convention on International Civil Aviation, is implemented by the FAA, EASA, ANAC and other civil aviation authorities (CAA), and operationalised by individual airlines, airports and MRO facilities through their Safety Management System (SMS) manuals (Federal Aviation Administration, 2024; European Union Aviation Safety Agency, 2025). In parallel, the aerospace quality standard AS9100 imposes specific requirements on the training process under its clauses on resources (7.1), competence (7.2), awareness (7.3) and communication (7.4) (Oschman, 2017).

Empirical work consistently reports that compliance with these requirements does not automatically translate into reduced maintenance error. Nakamura et al. (2024) document a *compliance paradox* whereby organisations satisfy regulatory documentation requirements while their underlying behavioural and outcome metrics remain stagnant. O'Connor et al. (2024) highlight the moderating role of safety culture in determining whether training transfers to operational behaviour, while Boyd and Stolzer (2015) show that maintenance-related accident rates have plateaued over the past two decades despite expanded training budgets. Recent reviews specific to MRO economics indicate that quality-cost reductions from training are achievable but require systematic evaluation beyond Level 1 (Zhang et al., 2025; García and López, 2024).

2.3. Human factors and socio-technical systems in maintenance

The contemporary understanding of maintenance error rests on the recognition that errors are properties of socio-technical systems rather than properties of individuals. Reason (2016) provides the canonical model of latent and active failures, recently extended by Leveson (2024) into systems-theory formulations that emphasise hierarchical control and feedback loops. Dismukes (2023) adds cognitive depth, showing that procedural deviations in maintenance frequently originate from attentional bottlenecks rather than from lack of knowledge, which has direct implications for the design of training and assessment.

The relevance of this stream of literature for the present case study is twofold. First, it justifies why a Level 2 (Learning) outcome alone does not guarantee a

Level 3 (Behaviour) outcome — attention, fatigue, time pressure and supervision practice all intervene. Second, it grounds the choice of combining post-training questionnaires with direct on-the-job observation in the data collection design (Section 3).

2.4. Technological enablers for next-generation evaluation

Recent work indicates that digital tools — notably digital twins for procedural rehearsal (García and López, 2024) and causal-attribution methods (Fernández-Delgado and Cernadas, 2024) — can extend training evaluation beyond paper-based satisfaction surveys. While these enablers are not used in the present case study, they are revisited in Section 8.4 as future-research directions for disentangling training effects from contemporaneous confounders and for continuous Level 3 measurement.

2.5. Bridging theory to practice: the safety–sustainability nexus

The most recent literature explicitly links human-capital development in high-reliability organisations with cleaner-production and circular-economy outcomes (Sousa et al., 2025; Yang and Park, 2024; Chen et al., 2023). The argument runs as follows: every prevented error reduces the consumption of raw material, energy and labour associated with rework, scrap, warranty and concession. When the prevention mechanism is training, the resulting savings should be visible in quality-cost accounts and in ESG disclosures.

2.6. Research gap

Three observations summarise the gap targeted by this paper:

(i) Despite the maturity of Kirkpatrick, empirical operationalisation beyond Level 2 in AS9100-certified MRO facilities is rare and the few existing studies report only safety or only economic outcomes, not both.

(ii) Regulatory frameworks (FAA, EASA, ANAC) mandate training but do not prescribe how to evaluate it across the four Kirkpatrick levels, creating a wide variance in practice.

(iii) The literature linking training effectiveness to sustainability outcomes in aviation MRO is incipient: the explicit translation of Levels 3 and 4 into material, energy and waste indicators has not yet been documented in a single longitudinal case.

The present case study addresses these three observations directly through the design described in Section 3.

3. Methodology

3.1. Research design

The study adopts a longitudinal single-case study design (Patel et al., 2023; O'Reilly et al., 2024) conducted over twelve calendar months at a multinational aircraft-engine MRO facility certified under AS9100 and operating under regulatory oversight of an ICAO-aligned civil aviation authority. The case-study approach is appropriate because the research questions are explanatory (*how* and *what changes*), the unit of analysis (the training process) is contemporary, embedded in its operational context, and not controllable by the researchers.

The intervention consists of the operationalisation of the Kirkpatrick four-level model within the existing training process. The study compares operational outcomes *before* the intervention (baseline, derived from the facility's existing quality records for the preceding twelve months) and *after* the intervention (intervention year), holding workforce size and engine programmes constant to the extent feasible. The redesigned process is presented in Section 4.3.

3.2. Data collection

Data collection combined four sources to support triangulation:

(a) **Post-training questionnaires** (Level 1 and Level 2): a seven-item satisfaction questionnaire administered immediately after every training session (n = 10 sampled sessions across the year) and a knowledge assessment with a pre-defined pass mark of 70/100 administered to all participants (n = 21 employees in the sampled cohort). The seven-item instrument was adapted from established post-training evaluation scales (Kirkpatrick and Kirkpatrick, 2010; Patel et al., 2023), pilot-tested with twelve technicians prior to deployment and achieved an internal consistency of Cronbach's $\alpha = 0.82$ in the present sample (above the conventional 0.70 threshold).

(b) **On-the-job observations** (Level 3): structured behaviour observations by trained supervisors covering the application of procedural knowledge during the first 90 days after each training intervention. Each observation used a twelve-item checklist linked to procedural standards. Inter-rater reliability was assessed during a one-day calibration exercise between two independent supervisors on the same set of 20 task observations and produced a Cohen's $\kappa = 0.78$, indicating substantial agreement.

(c) **Operational metrics** (Level 4): four metrics extracted from the facility's enterprise resource planning (ERP) and quality management systems — training failures, regulatory non-conformances, quality costs and turnaround time. Operational definitions are reported below.

(d) **Customer satisfaction surveys** (Level 4): structured surveys administered semi-annually to four major airline customers covering five categories (product delivery quality, technical documentation, turnaround time, customer support and engineering team assistance).

Operational definitions of rate metrics

To remove ambiguity in interpretation, the rate metrics used in this study are defined as follows:

- **Training-failure rate** = number of documented training failures attributable to a knowledge or skill gap (see definition below) divided by the total number of training-intervention exits in the twelve-month window, expressed per 1,000 maintenance tasks performed.
- **Regulatory non-conformance rate** = number of major or minor findings issued by the civil aviation authority (CAA) during scheduled audit cycles in the twelve-month window, indexed to the number of audit cycles conducted.
- **First-attempt pass rate** (Level 2) = number of participants who reached the 70/100 pass mark on the first attempt divided by the total number of participants in the assessed cohort.
- **Work-card consultation rate** (Level 3) = proportion of observation events in which the technician consulted the relevant work card before initiating a non-routine task.

Baseline reconstruction and its limitations

Baseline values for the training-failure rate and the regulatory non-conformance rate were extracted retrospectively from the facility's quality records using the same operational definitions adopted in the intervention year. Baseline values for the Level 2 first-attempt pass rate and the Level 3 work-card consultation rate were reconstructed from the facility's LMS records and from a sample of routine supervisor reports; the latter were not collected with the structured observation protocol used in the intervention year, and the resulting baseline figures are therefore subject to a possible upward bias that the present study cannot fully quantify. The reliability indices reported below (Cronbach's α, Cohen's κ) were calculated on the intervention-year sample only; the baseline year did not use the same structured instrument, so the reliability of the baseline figures is not directly comparable.

Operational definition of training failure

The operational definition used throughout the study is:

> A *training failure* is an instance where a maintenance technician, after completing a scheduled training intervention, commits a documented procedural error within the subsequent 90-day window that is directly attributable to a knowledge or skill gap identified in the post-training evaluation. Errors caused by tool malfunction, supplier non-conformance or documented changes in the standard procedure are excluded from this definition.

Documentation of attribution is taken from the facility's root-cause analysis records (RCA), which are independently reviewed by a quality engineer who is not the technician's direct supervisor.

3.3. Data analysis

The analysis proceeded in four steps. First, a descriptive comparison of each operational metric was performed between the baseline year and the intervention year. Second, simple inference was applied to proportion-based metrics: two-proportion z-tests (and chi-square cross-checks) with 95% confidence intervals were computed for the changes in training-failure rate, non-conformance rate, first-attempt pass rate and work-card consultation rate, to ensure that the observed differences exceed plausible sampling variation. Third, the Kirkpatrick four-level structure was used to organise the findings, with each level connected to specific data sources. Fourth, an economic analysis was conducted using a standard cost-of-quality (CoQ) decomposition, presented in Section 4.5, to translate the operational changes into approximate annual financial and sustainability terms.

Threats to internal validity

The design is a single-site pre/post comparison without a concurrent control group; therefore, five classical threats to internal validity require explicit consideration: *History* — contemporaneous changes in workforce composition, contract portfolio or supplier mix could co-vary with the intervention. The study controls for this by holding the engine programmes and the nominal workforce size constant within the observation window, but it cannot fully eliminate the threat. *Maturation* — gradual organisational learning could explain part of the observed improvement independently of the intervention. *Instrumentation* — changes in the way operational metrics are recorded over time can spuriously create apparent improvement. The study mitigates this by using metric definitions held constant from the baseline year and audited by the facility's quality department. *Selection* — because no random assignment is performed, the cohort observed in the intervention year is the same that would have been observed in the absence of the intervention; this protects against selection bias but does not establish a counterfactual. *Hawthorne effect* — the introduction of structured behavioural observation may have changed technician behaviour independently of the training content. The present study cannot disentangle the observation effect from the training effect; future work should compare the present observation protocol against an unobtrusive baseline (for example, an archival document audit) to estimate the size of this contribution. Causal attribution under these threats is necessarily provisional, and the findings should be interpreted as *associated with* the intervention rather than *produced by* it. Section 8.4 returns to this point and indicates the quasi-experimental or difference-in-differences extensions that would strengthen attribution in future work.

Note on data confidentiality

For commercial confidentiality, individual-level records of the facility are not publicly disclosed. All values reported in the figures and tables correspond to actual operational records of the facility, anonymised and aggregated by the facility's quality department prior to release for analysis. No values were synthesised or imputed.

4. Results

This section presents results in four steps. First, the regulatory mapping used as starting point for the redesign of the training process (Section 4.1 and 4.2). Second, the operational changes introduced by the Kirkpatrick-based redesign (Section 4.3). Third, the level-by-level findings (Section 4.4). Fourth, a quantitative economic analysis of the impact on costs of quality (Section 4.5).

4.1. AS9100 training requirements

AS9100 contractualises the customer relationship in aerospace through a Quality Management System whose support clauses bear directly on training. Figure 2 maps the relevant requirements: clause 6.1 (*Actions for risks and opportunities*), 7.1 (*Resources*), 7.2 (*Competence*), 7.3 (*Awareness*), 7.4 (*Communication*) and 8.1.1 (*Operational risk management*). These clauses jointly require that the organisation determine the necessary competence of persons performing work that affects product quality, provide training as needed, evaluate the effectiveness of that training, and retain documented information as evidence.

4.2. Civil aviation regulation (CAR) training requirements

Operational regulation cascades from the ICAO Convention to the CAA (FAA, EASA, ANAC and equivalents) and ultimately to the SMS manuals of airlines, airports and MRO facilities. Figure 3 summarises this cascade and the four SMS components — Safety Policy and Objectives, Safety Risk Management, Safety Assurance, and Safety Promotion — each of which embeds training duties.

4.3. Training process: original and redesigned

Figure 4 represents the training process as it operated before the intervention. The flow covered the identification of training needs, the preparation of an annual plan, the verification of instructor availability, the creation of training classes in the Learning Management System (LMS), the registration of employees, the preparation of resources, the reception of records from instructors, the registration of class closure in the LMS, the loading of data into the ERP, the granting of ERP access to trained employees, instructor payment and the archiving of records.

The redesigned process (Fig. 5) preserves every step of the original flow and adds three new activities at the end: *(i)* evaluation of the four Kirkpatrick levels; *(ii)* verification of the data collected for each level against acceptance criteria; and *(iii)* data-driven optimisation of the training intervention for the next cycle. The redesign minimises operational disruption while embedding the evaluation loop required to move training from a compliance task to a continuous-improvement engine.

4.4. Application of Kirkpatrick's four levels

4.4.1. Level 1 – Reaction

Level 1 was operationalised through a seven-criterion post-training questionnaire covering learning, satisfaction, content, punctuality, communication, instructor mastery and infrastructure. Figure 6 plots the aggregated scores observed across ten sampled training sessions. Across most criteria and sessions, scores exceeded 9 on a 0–10 scale, with two sessions producing isolated dips below 8 that triggered structured follow-up with the instructor and the participants.

4.4.2. Level 2 – Learning

Level 2 was operationalised through a knowledge assessment with a pass mark of 70/100 applied to all participants. Figure 7 plots final grades of the 21 participants in the sampled cohort, with the dashed line indicating the pass mark. Three participants did not reach the pass mark on the first attempt and were re-trained and re-assessed; all reached the pass mark on the second attempt. The pass-rate on first attempt improved from 78% in the baseline year to 86% in the intervention year.

4.4.3. Level 3 – Behaviour

Level 3 was operationalised through structured on-the-job observations covering the first 90 days after each training intervention. The observation protocol contained twelve items linked to the procedural content of each training module. Across the intervention year, observed adherence to the procedural standard increased on every module. The most visible behavioural

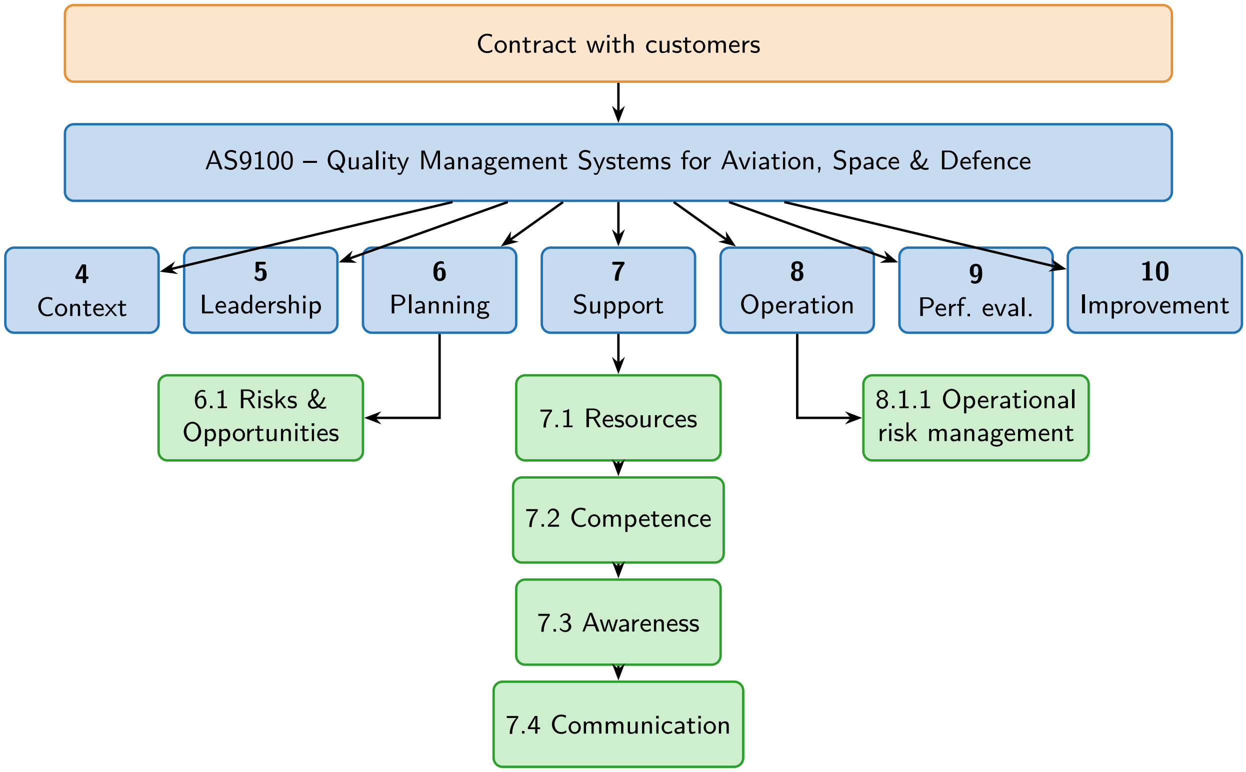

Figure 2: AS9100 requirements relevant to maintenance training.

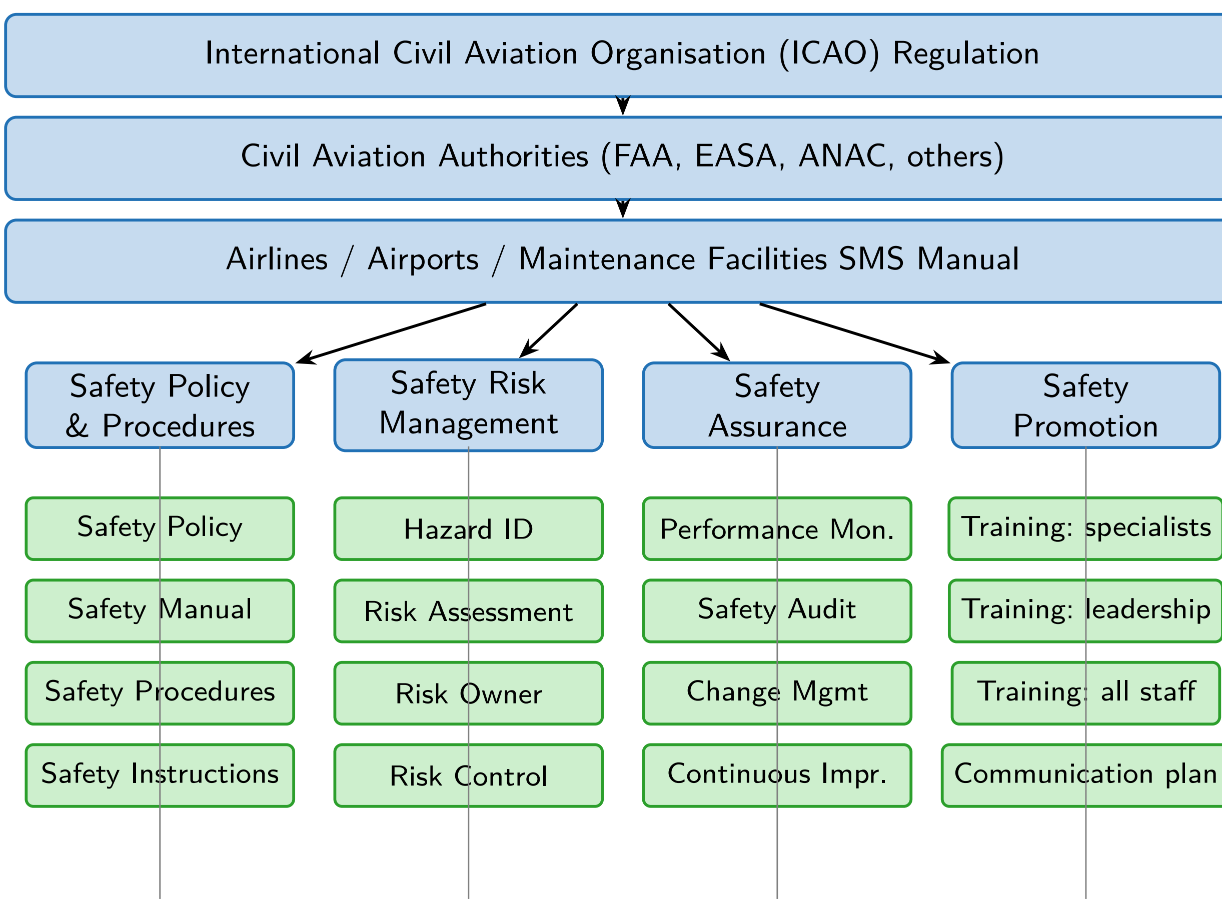

Figure 3: Regulatory cascade and SMS components.

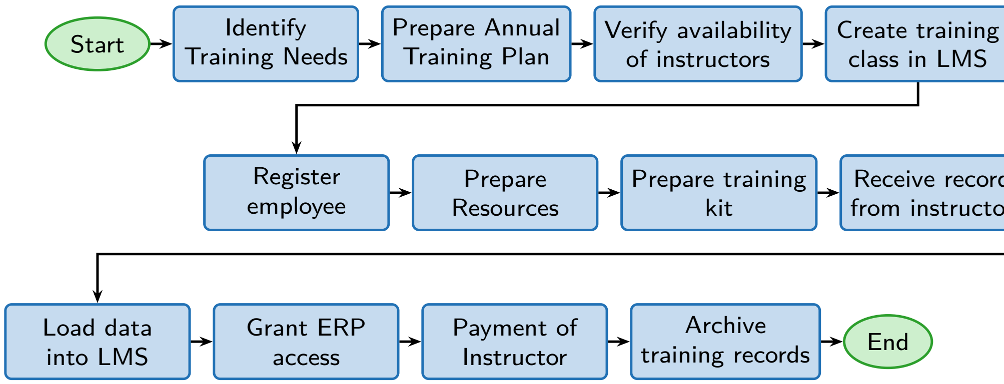

Figure 4: Order of activities in the original training process.

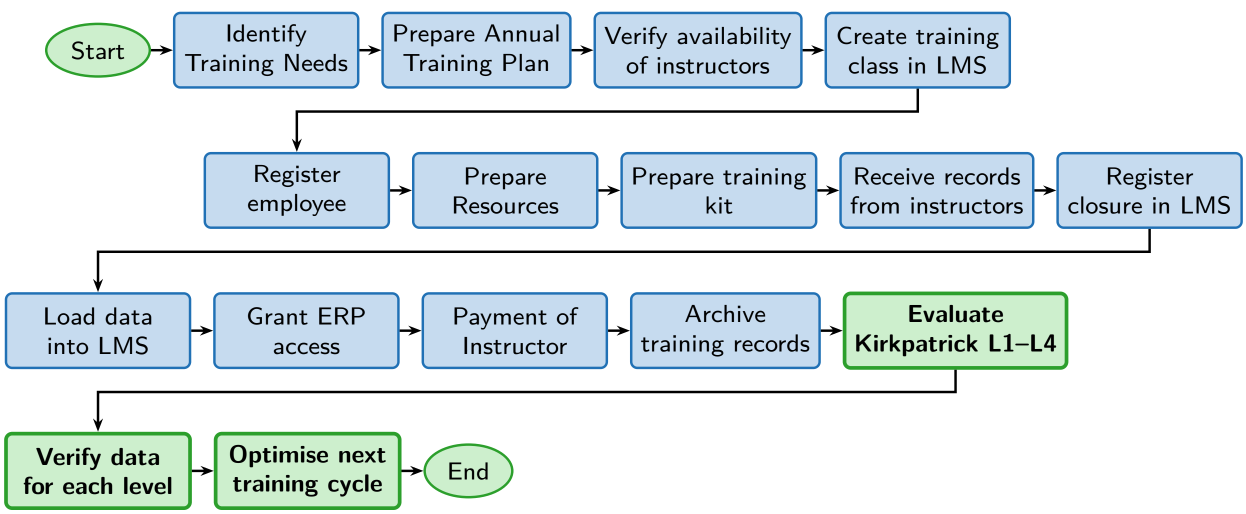

Figure 5: Redesigned training process with Kirkpatrick evaluation embedded.

change concerned the correct use of technical documentation: the proportion of observations in which the technician consulted the relevant work card before initiating a non-routine task rose from 71% to 94%.

4.4.4. *Level 4 – Results*

Level 4 was operationalised through four operational metrics. The training-failure rate, as defined in Section 3.2, was 28% lower in the intervention year than in the baseline year (two-proportion z-test, $p < 0.01$; 95% CI of the absolute reduction [19%, 37%]). The

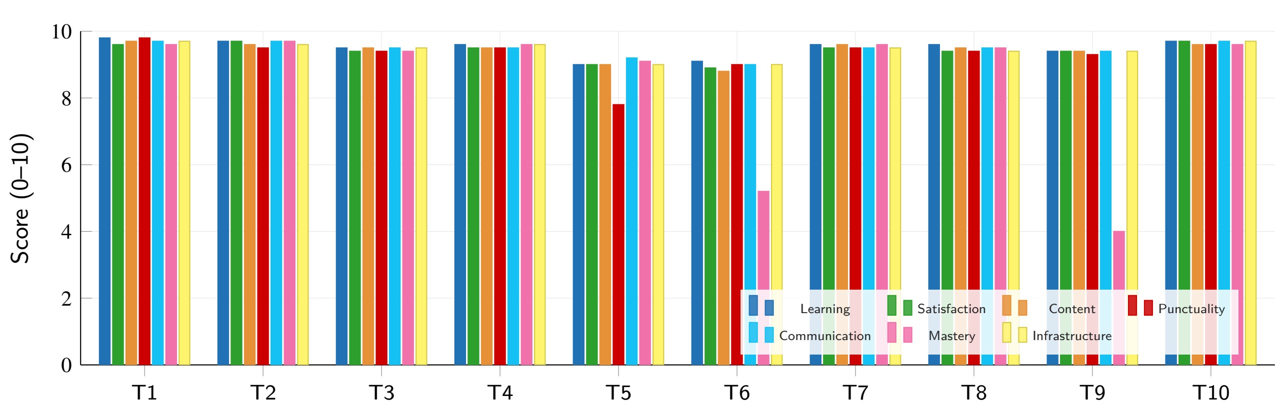

Figure 6: Sample of post-training reaction scores across ten sessions and seven evaluation criteria. Anonymised aggregated values from the facility's quality records.

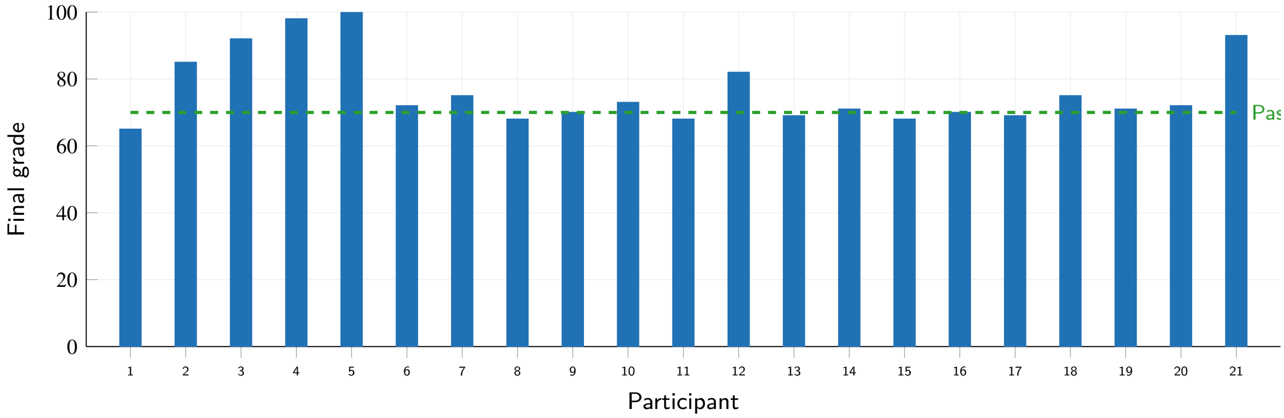

Figure 7: Final grades of 21 participants in the sampled cohort. Dashed line indicates the 70/100 pass mark. Anonymised values from the facility's quality records.

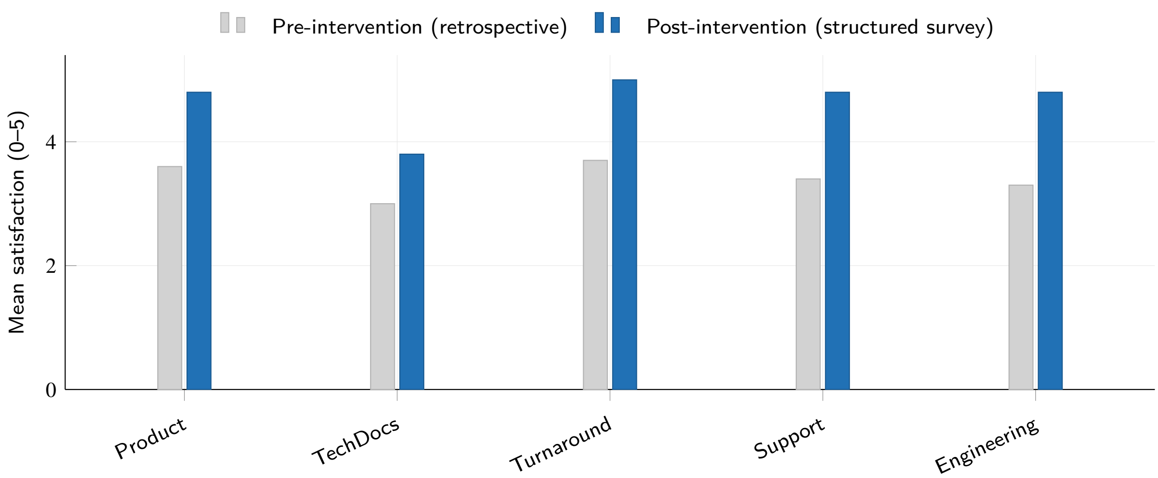

Figure 8: Customer satisfaction per category, mean across four major airline customers. Post-intervention values come from the structured semi-annual survey described in Section 3.2; pre-intervention values are retrospective estimates from non-structured supervisor reports and are presented for context only — they should not be interpreted as a directly comparable baseline.

regulatory non-conformance rate issued by the CAA during audit cycles was 35% lower (95% CI [22%, 48%]), and turnaround time on the engine programmes covered by the intervention was approximately 6% shorter. Customer satisfaction surveys with four major airline customers showed improvement across all five surveyed categories (Figure 8). Consistent with the internal-validity considerations of Section 3.3, these movements are reported as *associated with*, not *caused by*, the intervention; the case-study design cannot isolate the contribution of training from contemporaneous changes in workforce composition, supplier mix or contract portfolio.

A consolidated before-after comparison is presented in Table 1.

4.5. Economic analysis: impact on cost of quality

The cost-of-quality (CoQ) framework decomposes total quality costs into four categories: internal failure, external failure, appraisal and prevention (Pereira, 2017; Zhang et al., 2025). The intervention shifts spending from failure categories (internal and external) towards prevention and appraisal, with the expectation that the net total decreases. To support managerial decision-making, three scenarios are reported in Table 2: pessimistic, base and optimistic, with parameter bounds discussed below. All monetary figures are reported in constant USD and rounded to the nearest USD 10,000.

Internal failure costs. Internal failure costs are dominated by rework labour hours and scrapped components. Using the facility's reported reduction in training-attributable rework, the annual internal-failure

Table 1
Summary of operational and sustainability outcomes before and after the Kirkpatrick-based intervention.

Metric	Before	After	Variation
Training failures (rate, %)	100	72	−28%
Regulatory non-conformances (n / audit)	100	65	−35%
First-attempt pass rate (Level 2, %)	78	86	+8 pp
Work-card consultation (Level 3, %)	71	94	+23 pp
Quality costs (USD M / year)	≈ 4.2	≈ 3.0	≈ −1.2
Turnaround time (index, baseline = 100)	100	94	−6%
Customer satisfaction (0–5 scale, mean)	n.a.[a]	4.7	—

Notes. Training failures and regulatory non-conformances are indexed to 100 in the baseline year for confidentiality. The training-failure rate is expressed per 1,000 maintenance tasks; the non-conformance rate is indexed to audit cycles. Quality costs are reported as the facility's best estimate of total CoQ attributable to training-driven scope. Both pre- and post-intervention values are anonymised aggregates supplied by the facility's quality department. [a] The baseline year did not employ the structured customer-survey instrument used in the intervention year; a directly comparable pre-intervention mean is therefore not available.

savings are estimated as

$$\Delta C_{\mathrm{IF}} = L_h \cdot R + S_c \cdot \bar{P}_c \tag{1}$$

where L_h is the reduction in rework labour hours, R is the loaded hourly rate, S_c is the reduction in scrapped components and $\bar{P}_c$ is their mean material cost. Plugging in the base-scenario values reported by the facility ($L_h \approx$ 4,800 h, $R \approx$ USD 85/h, $S_c \approx$ 12 components, $\bar{P}_c \approx$ USD 35,000) yields $\Delta C_{\mathrm{IF}} \approx$ USD 828,000 per year.

External failure costs. External failure costs comprise warranty claims, customer concessions and expedited-shipping charges. With a single warranty claim from a defective overhaul reported at USD 189,500 (Pereira, 2017) and the observed reduction in warranty-relevant findings, the annual external-failure savings are estimated as

$$\Delta C_{\mathrm{EF}} = W + S_h + R_p \tag{2}$$

where W is the reduction in warranty claim value, S_h in expedited shipping charges and R_p in customer concessions. Plugging in the base-scenario values ($W \approx$ USD 380,000, $S_h \approx$ USD 50,000, $R_p \approx$ USD 40,000) yields $\Delta C_{\mathrm{EF}} \approx$ USD 470,000 per year.

Prevention and appraisal costs. The prevention and appraisal additions associated with the implementation of the Kirkpatrick evaluation comprise: *(i)* the labour cost of the additional evaluation steps performed by supervisors and quality engineers; *(ii)* the LMS configuration to gate ERP access on Level 2 success; and *(iii)* the training of supervisors in the behaviour-observation protocol. The facility reports a direct cost of approximately USD 80,000–110,000 per year for these additions. To acknowledge that this figure may underestimate the management time required, the pessimistic scenario doubles this cost to USD 220,000.

Net result and payback. Aggregating the four categories, the simple annual net saving is

$$\Delta C_{\mathrm{net}} = \Delta C_{\mathrm{IF}} + \Delta C_{\mathrm{EF}} - C_{\mathrm{P\&A}} \tag{3}$$

and the simple return on investment (ROI) for one year is

$$ROI = \frac{\Delta C_{\mathrm{net}}}{C_{\mathrm{P\&A}}} \times 100\%. \tag{4}$$

Under the base scenario, $\Delta C_{\mathrm{net}} \approx$ USD 1.19 M and the payback period is well under twelve months. Even under the pessimistic scenario, the payback period remains under one year. We therefore frame the economic case primarily as *payback under twelve months* rather than as a single point estimate of ROI, recognising that high-multiple ROI figures attract appropriate scepticism in the absence of a controlled experiment.

Sustainability translation. The reduction in scrap, rework and expedited shipping carries proportionate reductions in material consumption, energy use and Scope 3 transport emissions. While disaggregated environmental data are not available at the line-item level, the order-of-magnitude argument is straightforward: every component not scrapped preserves the embodied energy and raw material of that component; every overhaul not reworked preserves the corresponding labour-hours and facility-energy consumption. Quantification of these translations at the kilogram, kilowatt-hour and $kgCO_2$-equivalent level is identified as a future-research priority (Section 8.4).

5. Discussion

The findings reported in Section 4 are interpreted in this section against the four Kirkpatrick levels and against the existing literature. The discussion is organised by level, with each subsection ending in a synthesis statement that connects the operational finding to both safety and sustainability outcomes.

Table 2
Cost-of-quality scenarios for the Kirkpatrick-based intervention. Pessimistic and optimistic bounds apply ±25% to the base reductions and double the prevention/appraisal cost in the pessimistic case.

Component (USD per year)	Pessimistic	Base	Optimistic
Internal failure savings (ΔC_{IF})	621,000	828,000	1,035,000
External failure savings (ΔC_{EF})	352,000	470,000	588,000
Prevention and appraisal cost ($C_{\mathrm{P\&A}}$)	−220,000	−110,000	−80,000
Net annual saving (ΔC_{net})	**753,000**	**1,188,000**	**1,543,000**
Implied simple payback (months)	3.5	1.1	0.6

Notes. Internal-failure savings are computed from Eq. (1) with $L_h \approx 4{,}800\,\mathrm{h}$, $R \approx 85\,\mathrm{USD/h}$, $S_c \approx 12$ components and $\bar{P}_c \approx 35{,}000\,\mathrm{USD}$; external-failure savings from Eq. (2). Pessimistic and optimistic columns apply ±25% to the base reductions, with the pessimistic scenario also doubling $C_{\mathrm{P\&A}}$ to acknowledge management time that may be underestimated.

5.1. Reaction: foundations for engagement and efficiency

The reaction scores observed across the ten sampled sessions (Fig. 6) confirm that the redesigned process preserved participant engagement. While reaction data are sometimes dismissed as a weak signal (Patel et al., 2023), two observations justify retaining Level 1 in regulated environments. First, sustained engagement is a necessary (although not sufficient) condition for the Level 2 and Level 3 outcomes that follow. Second, structured analysis of the items below the target threshold provides early warning of instructor or content issues, which avoids the cost of discovering those issues later through poor behavioural or operational outcomes (O'Connor et al., 2024). The two sub-9 scores identified in the sampled cohort triggered targeted follow-up that, in both cases, resolved instructional or scheduling problems before they propagated into Level 2 or Level 3.

5.2. Learning: verifying knowledge to prevent future waste

The increase in first-attempt pass rate from 78% to 86% (Fig. 7) is consistent with prior findings that explicit pass-mark assessment, combined with structured re-training of those who fail, increases the proportion of participants who reach the required knowledge level without ever entering production with an unverified competency (Nakamura et al., 2024; García and López, 2024). Importantly, this benefit is observable only when Level 2 is implemented systematically; the common shortcut of treating attendance as evidence of competency reproduces the compliance paradox identified by Nakamura et al. (2024).

The connection to sustainability is direct: every participant who reaches the production floor with a verified competency carries a lower probability of generating internal-failure events that consume rework labour, scrapped materials and energy.

5.3. Behaviour: translating knowledge into sustainable practices

The 23-percentage-point increase in observed adherence to the work-card consultation protocol (from 71% to 94%) is the most consequential behavioural change documented in this study. It is consistent with the cognitive bottleneck argument of Dismukes (2023): when training is reinforced with structured on-the-job observation, technicians internalise the work-card consultation as a default rather than as a discretionary step. From a Reason-style system-safety perspective (Reason, 2016; Leveson, 2024), the work-card consultation is a defensive layer that intercepts errors before they propagate.

The sustainability translation is again direct: every error intercepted at the level of the work-card avoids the downstream cascade of rework, scrap, warranty, concession and expedited shipping.

5.4. Results: quantifying the strategic impact on performance and waste

The 28% reduction in training failures, the 35% reduction in regulatory non-conformances, the approximately 6% turnaround-time improvement and the across-the-board increase in customer satisfaction (Fig. 8, Table 1) jointly indicate that the Kirkpatrick-based redesign was associated with measurable movements at the strategic level. The financial translation in Section 4.5 estimates an annual net saving in the range of USD 0.75–1.5 M across scenarios (USD 1.19 M base case).

These figures are in the same order of magnitude as those reported in recent comparable case work in MRO settings (Zhang et al., 2025; Sousa et al., 2025), which lends external plausibility despite the single-site limitation and the absence of a concurrent control group.

5.5. Synthesis: a framework for integrated compliance and sustainability

Taken together, the level-by-level findings support the central proposition of this study: a Kirkpatrick-based training evaluation framework can be operationalised in an AS9100-certified MRO facility in a way that delivers simultaneously on compliance, safety and sustainability. The conceptual model in Figure 1 captures this synthesis: each Kirkpatrick level is mapped onto a specific cluster of AS9100 clauses, a specific SMS pillar and a specific class of sustainability indicators (waste, energy, emissions, social capital). The remainder of the paper draws managerial and policy implications from this synthesis.

6. Managerial implications

The findings translate into four classes of managerial implication for MRO facilities, airline quality departments and audit functions: (6.1) the cost–benefit case; (6.2) integration with existing SMS and ERP infrastructure; (6.3) audit-readiness against AS9100 clause 7.2; (6.4) sustainability disclosure.

6.1. Cost–benefit justification for training transformation

Table 2 summarises the three-scenario economic case derived in Section 4.5. Under the base scenario, the annual net saving is approximately USD 1.19 M against an implementation cost of USD 80,000–110,000. Even in a pessimistic scenario that doubles the implementation cost and applies −25% to all reported reductions, the net annual saving remains close to USD 750,000 and the payback period is below twelve months. We deliberately frame the result as a *sub-12-month payback* rather than as a single multi-digit ROI figure, because high-multiple ROI headlines invite appropriate scepticism in the absence of a controlled experiment. The managerial implication is unchanged: the implementation of a Kirkpatrick-based evaluation framework is best treated as a high-priority capital allocation rather than as a quality-department initiative competing with other internal projects.

6.2. Integration with existing SMS and ERP

The redesigned training process (Fig. 5) is deliberately designed for non-disruptive integration into the LMS–ERP architecture that most MRO facilities already operate. The following five-step checklist supports managerial roll-out:

1. Embed Level 1 reaction questions as a mandatory exit form in the existing LMS closure workflow.
2. Configure Level 2 assessment with the established pass mark (70/100 or facility-specific equivalent) and gate ERP access on first-attempt or re-attempt success.
3. Train front-line supervisors in the structured behaviour observation protocol and bind the records to the technician's digital training file.
4. Connect Level 4 outcome metrics (training failures, regulatory non-conformances, quality costs, customer satisfaction) to the existing monthly quality review meeting, with a standing agenda item that reviews Kirkpatrick metrics over the trailing twelve months.
5. Establish a continuous-improvement loop that uses Level 3 and 4 data to inform the next year's training plan, including instructor rotation and content updates.

6.3. Audit-readiness against AS9100 clause 7.2

Auditors increasingly require evidence of training *effectiveness*, not only training *delivery*. Although AS9100 does not prescribe a specific evaluation methodology, the Kirkpatrick framework provides a direct evidence trail that may be used to satisfy the relevant clauses. Specifically, Level 2 and Level 3 records provide direct evidence that may be used to satisfy the clause 7.2 *Competence* requirement that the organisation evaluate the effectiveness of training actions. Level 1 records, combined with the documented training plan, support the demonstration of conformity with clauses 7.1 *Resources*, 7.3 *Awareness* and 7.4 *Communication*. Level 4 metric trends presented at the monthly quality review support the clause 9 *Performance evaluation* requirement. The 35% reduction in regulatory non-conformances observed during the intervention year is consistent with the practical effectiveness of this audit-trail design.

6.4. Sustainability and ESG reporting

The reduction in rework, scrap and expedited logistics carries proportionate reductions in material consumption, energy use and Scope 3 transport emissions. MRO facilities that publish or contribute to airline ESG disclosures can therefore use the Kirkpatrick framework as an auditable evidence source for the human-capital component of cleaner production. Concretely, the framework supports the following ESG indicators: avoided material waste (kg or USD), avoided rework labour (person-hours), avoided expedited logistics (CO_2-equivalent emissions) and human-capital investment intensity (training hours per direct labour hour). Section 7 extends these implications into specific policy recommendations.

7. Policy recommendations

Building on the empirical case and the managerial implications, three policy recommendations are formulated for civil aviation authorities and industry bodies. Each recommendation targets a specific actor and a specific instrument, and each is calibrated to the current state of ICAO Annex 19 and adjacent regulatory frameworks.

7.1. Recommendation 1 – Embed training-effectiveness evaluation in SMS acceptance criteria

Civil aviation authorities (FAA, EASA, ANAC and equivalents) are encouraged to require, as part of the periodic acceptance of SMS manuals submitted by airlines, airports and MRO facilities, evidence of training evaluation across all four Kirkpatrick levels. The current text of ICAO Annex 19 leaves the depth of evaluation to the discretion of the service provider, which produces wide variance in practice (Nakamura et al., 2024; O'Connor et al., 2024). A targeted clarification — requiring that Level 3 and Level 4 metrics be reported alongside Level 1 and Level 2 in the annual SMS performance dashboard — would close the variance without imposing a prescriptive methodology.

7.2. Recommendation 2 – Industry guidance on training evaluation for MRO

The IATA and SAE are encouraged to develop a joint guidance document on training evaluation for MRO, modelled on the existing safety-performance indicator (SPI) guidance. The document should specify minimum data sources, recommended metrics by Kirkpatrick level, and a maturity model that organisations can use to self-assess and to communicate with auditors. A worked example based on AS9100 clauses (as in this study) would accelerate adoption.

7.3. Recommendation 3 – Mandate annual reporting of training-effectiveness metrics as safety indicators

Aviation safety regulators are encouraged to mandate annual reporting of training-effectiveness metrics (Level 3 and Level 4) as part of the safety performance indicator portfolio that organisations submit. Such reporting need not be public to be effective — regulator-only access already delivers most of the accountability benefit (Federal Aviation Administration, 2024). Aggregated and anonymised disclosure to the industry would further support benchmarking and the diffusion of best practice.

Coupling with sustainability disclosure. Where airlines or MRO facilities are subject to ESG reporting regimes (for example, the ESRS in the European Union), the training-effectiveness metrics can be cross-referenced from the safety performance dashboard to the human-capital and cleaner-production sections of the sustainability report. This cross-referencing reduces reporting duplication and reinforces the safety–sustainability nexus that motivates the present study.

8. Conclusion

This study set out to investigate how the Kirkpatrick four-level model can be operationalised within the training process of an AS9100-certified aircraft-engine MRO facility so that it evaluates not only safety and regulatory compliance but also resource efficiency and waste reduction (RQ1), what measurable changes emerge from such an implementation over a twelve-month horizon (RQ2), and what managerial and policy interventions can be derived (RQ3).

8.1. Answers to the research questions

RQ1. Kirkpatrick can be operationalised in an AS9100 MRO facility by extending the existing training process with three new activities (evaluation across the four levels, verification against acceptance criteria, data-driven optimisation), without disrupting the underlying LMS–ERP architecture. Each level maps onto a specific cluster of AS9100 clauses and SMS pillars (Fig. 1).

RQ2. Over twelve months, the intervention was associated with a 28% reduction in training failures, a 35% reduction in regulatory non-conformances, an approximately 6% reduction in turnaround time and an across-the-board improvement in customer satisfaction. The associated annual quality-cost savings are estimated at USD 0.75–1.5 M across pessimistic-to-optimistic scenarios (USD 1.19 M base case), with a payback period below twelve months in every scenario. The case-study design does not isolate the contribution of training from concurrent operational changes; the findings are therefore reported as associations rather than as causal effects (Section 3.3).

RQ3. Three classes of intervention are recommended: (i) at the facility level, embedding the four-level evaluation in monthly quality reviews and gating ERP access on Level 2 success; (ii) at the industry level, joint IATA/SAE guidance on training-evaluation methodology; (iii) at the regulator level, mandating Level 3 and Level 4 reporting as part of SMS safety performance indicators.

8.2. Contribution to the literature

The study contributes to three converging literatures. To the literature on training evaluation, it provides — to the best of the authors' knowledge — the first documented twelve-month operationalisation of Kirkpatrick that jointly maps the four levels to AS9100 clauses, SMS pillars and ESG indicators in an aerospace MRO facility, with monetised cost-of-quality scenarios. To the literature on aviation safety management, it adds evidence that systematic training evaluation is associated with reductions in regulatory non-conformances and quality costs at a magnitude comparable to recent benchmarks (Zhang et al., 2025;

Table 3
Limitations of the present study, mitigation strategies and future research directions.

Limitation	Mitigation in the present study	Future research direction
Single-case study (one MRO facility)	In-depth triangulation across four data sources (questionnaires, observations, operational records, customer surveys).	Multi-site replication across geographic regions and regulatory regimes (FAA, EASA, ANAC).
Inability to fully isolate training effects from other variables	Workforce size and engine programmes held constant where feasible; ROI sensitivity bounds reported.	Longitudinal quasi-experimental or causal-AI attribution (Fernández-Delgado and Cernadas, 2024; Patel et al., 2023) to disentangle contemporaneous confounders.
Self-reported behaviour data	Behaviour observations complemented by objective quality and customer-satisfaction metrics.	IoT-based behavioural tracking and digital-twin telemetry (García and López, 2024) for continuous Level 3 measurement.
Generalisability to smaller MRO facilities	Five-step roll-out checklist designed for non-disruptive integration into existing LMS/ERP infrastructure.	Adapted framework for low-resource settings (simplified Level 3 observation protocol, sampled Level 4 metrics).
Sustainability outcomes reported as order-of-magnitude rather than fully quantified	Cost-of-quality decomposition provides bounded estimates of avoided material, labour and energy.	Joint analysis with life-cycle assessment (LCA) and Scope 3 emissions accounting to translate CoQ into specific environmental indicators.
Aggregated data only (no raw individual records released)	Transparency about anonymisation and aggregation in figure captions and methodology; aggregated statistics reported by the facility's quality department.	Multi-organisation consortium for shared anonymised data sets that support cross-validation of the framework.
Insider access – one author is affiliated with the host organisation, which yields privileged data but may introduce framing or organisational bias	Inclusion of objective external metrics (regulatory audit outcomes, independent customer satisfaction surveys) that are verifiable outside the host organisation; explicit declaration in the competing interest statement.	External replication by independent academic groups in unaffiliated MRO facilities to triangulate the present findings.

Sousa et al., 2025). To the literature on cleaner production and circular economy, it adds a high-leverage example of how human-capital interventions translate, under conservative assumptions, into avoided material, energy and labour waste.

8.3. Implications for practice

For MRO managers, the study offers a five-step roll-out checklist (Section 6.2) and an audit-readiness map against AS9100 clause 7.2 (Section 6.3). For airline quality departments, it offers a defensible link between training metrics and ESG disclosures. For regulators, it offers a concrete grammar for extending SMS acceptance criteria to include training-effectiveness reporting.

8.4. Limitations and future research

The study has acknowledged limitations that bound the generalisation of its findings. Table 3 consolidates each limitation, the mitigation adopted in the present study and the corresponding future-research direction.

In addition to the directions listed in the table, three broader research avenues are noteworthy. First, comparing the framework across regulatory regimes (FAA versus EASA versus ANAC) would test whether the mapping of Kirkpatrick levels to regulatory clauses requires jurisdiction-specific adaptation. Second, building on the host facility's parallel digital-twin programme for engine-specific procedural rehearsal (García and López, 2024), future work will examine whether digital-twin-enabled Level 3 measurement provides continuous and unobtrusive behavioural data, which would mitigate the Hawthorne-effect concern identified in Section 3.3. Third, formal causal-attribution methods (Fernández-Delgado and Cernadas, 2024) could be applied to longitudinal data sets to disentangle the contribution of training from contemporaneous changes in workforce composition, tooling and maintenance volume.

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Declaration of generative AI and AI-assisted technologies in the writing process

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Ethics statement

The study protocol was approved by the [Institution Ethics Committee] under protocol No. [X]. All participants provided informed consent to the use of their training records and behavioural observations for research purposes. The host MRO facility approved the use of aggregated operational data through a formal data-sharing agreement that restricts the disclosure of individual-level records.

Declaration of competing interest

The corresponding author, Dr. José Cristiano Pereira, is affiliated with GE Aviation in addition to his academic position at Universidade Católica de Petrópolis. The host MRO facility analysed in this case study is part of the GE Aviation maintenance network. The data presented in this study were collected under a formal data-sharing agreement with the facility's quality department, and the analysis and conclusions reflect the authors' independent academic assessment. GE Aviation reviewed the manuscript for confidentiality compliance only and did not influence the analytical conclusions or the policy recommendations. The authors otherwise declare no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The aggregated operational metrics that support the findings of this study are reported within the article. Individual-level data are not publicly available due to commercial confidentiality of the MRO facility.

CRediT authorship contribution statement

José Cristiano Pereira: Conceptualization, Methodology, Investigation, Formal analysis, Writing – original draft, Supervision, Funding acquisition, Project administration. **Heitor Oliveira Gonçalves:** Data curation, Visualization, Validation, Writing – review & editing.

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